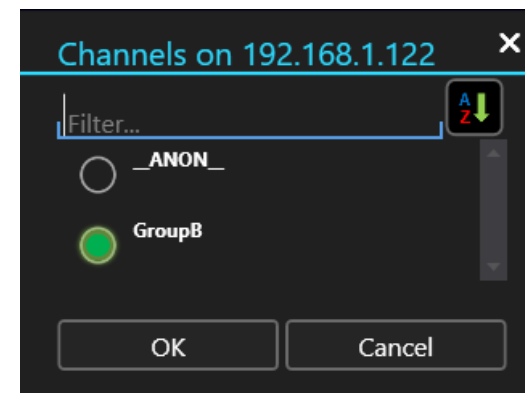
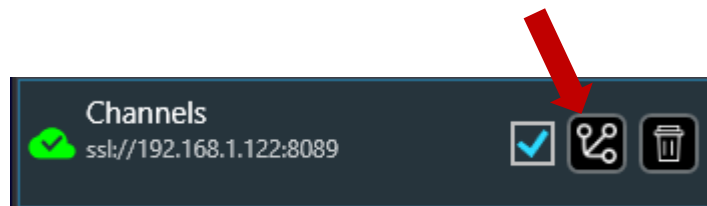


TAK Server - Channels

TAK Server - Intro to Channels

- Channels allow the user to pick which of their sharing groups are active at a given point in time.
- The user's selection affects all information shared on both the messaging and API ports.
- TAK Server admins can migrate from complex setups with many filter groups and a large network footprint, to a consolidated setup with a single messaging and API port.



TAK Server - Channels

- Channels build on the existing Certificate Enrollment and x509 Authentication capabilities in TAK.
- The CN (Common Name) of the certificate is set to the user's name, which the x509 Authentication extracts and performs the group lookup using the configured authenticator(File or LDAP).
- The groups returned will always provide the authoritative ground truth for the user's group membership.
- This additional step will allow the end user to select a subset of these groups to use for the current authentication request.

TAK Server - Channels Implementation

- Channels are disabled by default and needs to be enabled within the core configuration file.
- While the TAK Server is running, and you have an open terminal SSH'd into it do the following:
 - Change directory:
 - **cd /opt/tak**
 - Change to the tak user:
 - **sudo su tak**
 - Edit the CoreConfig.xml file with a text editor:
 - **vi CoreConfig.xml**

TAK Server - Channels Implementation

- Within the CoreConfig file find the following line:

```
<auth x509groups="true" x509addAnonymous="false" x509checkRevocation="true">  
  <File location="UserAuthenticationFile.xml"/>  
</auth>
```

- Add the following to the auth element:
 - **x509useGroupCache="true"**
 - If done properly it should look like the following:

```
<auth x509groups="true" x509addAnonymous="false" x509checkRevocation="true" x509useGroupCache="true">  
  <File location="UserAuthenticationFile.xml"/>  
</auth>  
<submission ignoreStaleMessages="false" validateXml="false"/>
```

- Note: that if you are using another file service, such as LDAP, the body of the auth element will be different.
- Exit the tak user:
 - **exit**
- Restart the TAK Server
 - **sudo systemctl restart takserver**

TAK Server - Channels Implementation

- To apply the channel configuration to the end-user we need to create a Device Profile, which allows you to upload data packages that can be applied to a device or group.
- Go to the Device Profiles section in the admin dashboard:
 - Hamburger menu → Administrative → Device Profiles
- To create a new profile:
 - Add a name for your profile in the section “Set Profile Name”
 - Select “Create”

Device Profile Manager

Name	Type	Tool	Groups	Active	
Testing	Enrollment		["GroupA","GroupB","__ANON__"]	true	Edit Send Download Delete

Channels

Create

TAK Server - Channels Implementation

- Download the following Channel Configuration data package file:
 - <https://mytecknet.com/content/files/2023/08/channels.zip>
- Within the new profile select the following:
 - Type: Apply on Enrollment
 - Apply on Enrollment: all new clients that connect using certificate auto-enrollment will have this device profile applied.
 - Apply on Connection : all clients that connect will have the device profile applied even if the client has already connected.
 - Groups: You can select which groups it applied to or use the “Select All” option.
 - Browse: Select the channel file you downloaded.
 - Select “Upload File”

Device Profile Manager / Channels

Type

- ☒ Apply on Enrollment
- ☐ Apply on Connection
- ☐ Tool

Tool (optional)

Groups

[Select All](#)

- ☐ GroupA
- ☒ GroupB
- ☒ __ANON__

Files

Name

No file selected.

TAK Server - Channels Implementation

- When you are done, select the “Device Profile Manager” parent directory to be taken back to the Device Profiles section.
- From here you can view and manage your profiles:
 - You have a couple options for the profiles to edit, send, download, or delete them.
 - One option to note is the “Send” option which allows you to send the device profile to a specific user.

Device Profile Manager

Name	Type ▼	Tool	Groups	Active	
Channels	Enrollment		["GroupB","__ANON__"]	true	Edit Send Download Delete
Testing	Enrollment		["GroupA","GroupB","__ANON__"]	true	Edit Send Download Delete

[Create](#)

TAK Server - Channels Integration

- Once connected to the TAK Server the user can select which groups to enable by selecting the following icon from the “Manage Server Connections” section:

ATAK



WinTAK

